

Practical work no 5

* Given an undirected graph, find a covering of edges by vertices with no more than twice the optimal number of vertices in $O(n+m)$ time.

1 a5_1.txt

5 6

0 1 1

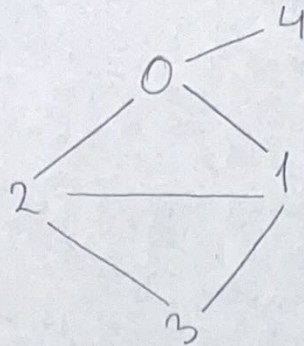
0 2 1

1 2 1

1 3 1

2 3 1

0 4 1



a) Approximation algorithm

edges = set of tuples

vertices = the covering of edges by vertices

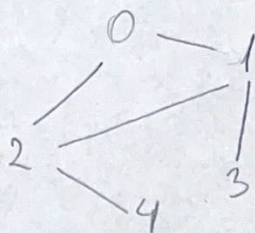
	edges	selected edge	vertices
initialization	$\{(0,1)(0,2)(1,2)(1,3)(2,3)(0,4)\}$	-	$\{\}$
iter 1	$\{(2,3)\}$	$(0,1)$	$\{0,1\}$
iter 2	$\{\}$	$(2,3)$	$\{0,1,2,3\}$

b) Greedy

	edges	degree	best vertex	vertices
initialization	$\{(0,1)(0,2)(1,2)(1,3)(2,3)(0,4)\}$	$\{\}$	-	$\{\}$
iter 1	$\{(1,2)(1,3)(2,3)\}$	$\deg[0]=3$ $\deg[1]=3$ $\deg[2]=3$ $\deg[3]=2$ $\deg[4]=1$	0	$\{0\}$
iter 2	$\{(2,3)\}$	$\deg[1]=2$ $\deg[2]=2$ $\deg[3]=2$	1	$\{0,1\}$
iter 3	$\{\}$	$\deg[2]=1$ $\deg[3]=1$	2	$\{0,1,2\}$

2. a5-2.txt

55
0 1 1
0 2 1
1 2 1
1 3 1
2 4 1



a) Approximation algorithm

	edges	selected edge	vertices
initialisation	$\{(0,1)(0,2)(1,2)(1,3)(2,4)\}$	-	$\{\}$
iteration 1	$\{(2,4)\}$	$(0,1)$	$\{0,1\}$
iteration 2	$\{\}$	$(2,4)$	$\{0,1,2,4\}$

b) Greedy

	edges	degree	best-vertices	vertices
initialisation	$\{(0,1)(0,2)(1,2)(1,3)(2,4)\}$	-	-	$\{\}$
iteration 1	$\{(0,2)(2,4)\}$	$\deg[0]=2$ $\deg[1]=3$ $\deg[2]=3$ $\deg[3]=1$ $\deg[4]=1$	1	$\{1\}$
iteration 2	$\{\}$	$\deg[0]=1$ $\deg[2]=2$ $\deg[4]=1$	2	$\{1,2\}$